

DARK VESSEL DETECTION & MARITIME INTELLIGENCE SYSTEM

Full Project Documentation

Development Direction Path for Teammates

Subject	Python for Data Science
Domain	Maritime Surveillance — Remote Sensing + AI
Core Technology	SAR + AIS Fusion, Fine-tuned VLM, Agentic Pipeline
Publication Target	IEEE Transactions on Geoscience and Remote Sensing
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1. Problem Statement

1.1 What Is a Dark Vessel?

Every commercial vessel above a certain size is legally required to continuously broadcast its identity and position via AIS — Automatic Identification System. This broadcast contains the vessel's name, IMO number, position, speed, heading, and destination, updated every few seconds.

Vessels engaged in illegal activity — sanctions evasion, unauthorized fishing inside a nation's Exclusive Economic Zone, arms or drug smuggling, human trafficking — deliberately disable their AIS transponders to avoid detection. A vessel operating without an active AIS signal is called a dark vessel.

1.2 The Scale of the Problem

- Global Fishing Watch estimates 25% of large vessels go dark in suspicious circumstances at some point
- The Arabian Sea and Bay of Bengal — India's primary maritime zones — are among the most active corridors for sanctions evasion
- Iran sanctions evasion via Gulf of Oman → Arabian Sea is a documented and ongoing pattern
- Illegal foreign fishing inside India's 2.37 million sq km EEZ causes an estimated \$3.6 billion annual loss
- Ship-to-ship transfer of sanctioned cargo happens in international waters specifically to avoid port state control

1.3 Why the Current Process Fails

Identifying dark vessels today requires a maritime intelligence analyst to:

- Manually download satellite imagery for a given ocean area and time window
- Manually cross-reference each ship-like target with AIS broadcast records
- Manually look up vessel registries, sanctions lists, and ownership databases
- Manually write an investigation report and prediction assessment

Per event this takes hours to days of expert labor. At the scale of global ocean traffic this is operationally impossible — most dark vessel events go undetected and uninvestigated.

1.4 India-Specific Context

India's Information Fusion Centre — Indian Ocean Region (IFC-IOR), established in 2018, exists specifically to monitor maritime domain awareness across the Indian Ocean. ISRO operates SAR satellites — RISAT-2B, RISAT-2BR1, RISAT-2BR2 — for exactly this maritime monitoring mission. Yet no automated intelligence generation system exists to support these institutions. Our system is designed to directly complement this existing national infrastructure.

2. What the System Does

2.1 System Input

- Coordinates of an ocean area of interest OR an uploaded SAR image
- Time window of interest
- Everything else is handled automatically by the agentic pipeline — no manual steps

2.2 Three Layers of Output

The system produces three layers of output in a single pipeline run. This is critical — the system does not stop at detection or at report generation. It goes through to predictive intelligence.

Layer 1 — Detection

SAR imagery is processed to detect all vessel-like targets. Each detected target is cross-referenced against AIS records for the same area and time. Targets present in SAR but absent in AIS are flagged as dark vessels. This is the sensing layer.

Layer 2 — Investigation Report

For each dark vessel: estimated vessel dimensions from SAR signature, probable vessel identity from registry cross-matching, sanctions status check, behavioral pattern history, and determination of what regulation is likely being violated. This is the analysis layer.

Layer 3 — Predictive Intelligence

This is what makes the system operationally useful rather than just informational. The system outputs:

- Next location prediction — where will this vessel be in 6, 12, 24 hours based on heading, speed, and known routes
- Activity prediction — what is this vessel about to do based on trajectory and behavioral pattern
- EEZ threat assessment — will this vessel enter Indian territorial waters and when
- Intercept window — how long until the vessel is out of practical intercept range
- Recurrence prediction — if this vessel or profile has appeared before, when will it appear again

2.3 Sample System Output

```
MARITIME INTELLIGENCE REPORT
SAR Pass: Sentinel-1A | 14 Aug 2024 | 03:42 UTC
Area: Arabian Sea - 19.3N, 62.1E

DARK VESSEL DETECTED - No AIS correlation

VESSEL CHARACTERISTICS (SAR-derived):
  Estimated length : 165-185m
  Estimated beam   : 22-26m
  Vessel class     : Bulk carrier or tanker
  Heading          : 247 degrees westbound
  Speed estimate   : 8-10 knots (SAR Doppler)

AIS STATUS:
  Last known signal : Gulf of Oman, 31 hours prior
  Gap duration      : 31 hours - deliberate disabling likely
```

IDENTITY ANALYSIS:

Top candidate: MV [Vessel Name] – IMO 9XXXXXXX
Flag: Panama | Owner: [Entity] on OFAC SDN list since March 2024
Confidence: HIGH – dimension match + route plausibility

PREDICTIVE INTELLIGENCE:

+6h position : 18.9N, 60.8E
+12h position : 18.5N, 59.4E
+24h position : 17.8N, 56.7E

Activity prediction: Track passes through known STS transfer zone at 18.5N, 61.2E – ETA 4.2 hours
Probability of ship-to-ship transfer: HIGH
Basis: 4 prior confirmed STS events match this approach vector

EEZ entry: Projected at 18.2N if course maintained – ETA 31h
Intercept window: 3.5 hours before vessel enters deep water zone

RECURRENCE PATTERN:

Matching profile detected dark: 43, 87, 134 days ago
Interval: ~44 days | Next predicted transit: ~44 days from today

RECOMMENDED ACTION:

Alert Indian Coast Guard Western Command – immediate
File report with IFC-IOR
Cross-reference with INTERPOL vessel watchlist

3. Data Sources

All data sources used in this project are free and publicly accessible. No proprietary data subscription is required for the academic prototype.

3.1 SAR Satellite Imagery

Source	Band	Resolution	Revisit	Archive	Access
Sentinel-1 (ESA)	C-band	10m (GRD)	6 days	Since 2014	Free — Copernicus Hub
RISAT-2B (ISRO)	X-band	1m	Sub-daily	Limited	NRSC portal — research
ALOS-2 PALSAR (JAXA)	L-band	3-10m	14 days	Since 2014	Research access

Sentinel-1 is the primary source. It provides complete global ocean coverage every 6 days, completely free, via the Copernicus Open Access Hub. The Python library `sentinelsat` enables automated programmatic download. SAR works at night and through clouds — it emits its own microwave pulse and reads the reflection, making it fundamentally different from optical satellites.

3.2 AIS Vessel Tracking Data

Source	What It Contains	Size	Access
Global Fishing Watch	AIS records + dark vessel events labeled	300M+ records	Free research API
NOAA AIS Archive	Historical AIS, US + international waters	Large	Free download
MarineTraffic	Global vessel position history	Complete	Research access
OpenSky Network	Vessel AIS — maritime extension	Growing	Free API

3.3 Vessel Registry and Sanctions

Source	What It Contains	Access
IMO GISIS	Official global vessel registry — all IMO-registered ships	Free — searchable
OFAC SDN List	US sanctions list — entities, vessels, updated frequently	Free API + XML download
UN Consolidated Sanctions	UN Security Council sanctions list	Free XML download
EU Sanctions Vessel List	European Union sanctioned vessels	Free download
Equasis	Vessel history, ownership chains, flag history	Free registration

3.4 Training Datasets for Fine-Tuning

Dataset	What It Contains	Size	Use
SAR-Ship	Labeled ship chip images in SAR	39,729 instances	Stage 1 detection training

Dataset	What It Contains	Size	Use
HRSID	High-resolution SAR ship images	5,604 images	Stage 1 — high-res training
OpenSARShip	Ships with matched AIS records	11,346 chips	AIS correlation training
SSDD	SAR ship detection benchmark	1,160 images	Evaluation benchmark
UNODC Reports	Maritime crime enforcement cases	500+ documents	Report generation training
OFAC Enforcement	Vessel-level penalty orders	Hundreds	Investigation narrative pairs
Coast Guard Press Releases	Interception events with vessel details	Ongoing	Labeled dark vessel cases

Training pairs for the report generation stage are constructed by parsing UNODC, OFAC, and coast guard documents into structured (input features → investigation narrative) pairs. Estimated constructable pairs: 300-500 high quality labeled examples — sufficient for QLoRA fine-tuning based on published literature.

4. Model Selection and Fine-Tuning

4.1 Why Fine-Tuning Is Necessary

A general vision-language model cannot perform this task out of the box because:

- SAR imagery looks nothing like optical photography — pixel brightness represents radar backscatter, not reflected light. Models trained on natural images fail on SAR.
- Vessel classification from SAR backscatter signatures requires domain knowledge — different vessel types (tanker, bulk carrier, container ship, submarine) have specific radar cross-section patterns the model must learn.
- AIS gap semantics — what a 31-hour gap in a specific corridor means operationally — is not in general pre-training data.
- Investigation reports must reference OFAC regulations, IMO conventions, and Indian Coast Guard procedures in correct legal language — domain-specific fine-tuning is what teaches this.

4.2 Base Model Options

Model	Parameters	Image Support	License	Recommended For
LLaVA-1.6 (Mistral-7B)	7B	Yes — multi-resolution	Apache 2.0	Primary choice — best QLoRA support
InternVL2-8B	8B	Yes — high resolution	MIT	Strong alternative — better remote sensing baseline
Qwen2-VL-7B	7B	Yes — dynamic resolution	Apache 2.0	Good — strong instruction following
CogVLM2	19B	Yes	CogVLM License	If compute allows — stronger reasoning

Primary recommendation: LLaVA-1.6 with Mistral-7B backbone. Best documented QLoRA fine-tuning pipeline, Apache 2.0 license, fits on single A100 40GB, and has published benchmarks on remote sensing tasks.

4.3 Fine-Tuning Method — QLoRA

QLoRA — Quantized Low-Rank Adaptation — is the fine-tuning approach for both stages.

- Model weights are quantized to 4-bit — reduces memory requirement from ~28GB to ~10GB for a 7B model
- Small LoRA adapter layers are added on top — only 1-2% of total parameters are actually trained
- Result: full domain adaptation without retraining billions of parameters
- Training time: 6-8 hours per stage on a single A100 40GB GPU (Google Colab Pro or university cluster)
- Library: HuggingFace PEFT + bitsandbytes + transformers

4.4 Two-Stage Training

Stage 1 — SAR Ship Detection and Classification

- Input: SAR image chip (single ship target)
- Output: vessel class (tanker / bulk carrier / container / fishing / unknown), dimension estimate, confidence score
- Dataset: SAR-Ship + HRSID + OpenSARShip — combined ~50,000 labeled examples

- Objective: model learns SAR backscatter signatures for different vessel types

Stage 2 — Investigation Report and Predictive Intelligence Generation

- Input: structured JSON of SAR detection output + AIS gap data + registry cross-reference results
- Output: full investigation report with predictive intelligence section
- Dataset: 300-500 constructed pairs from UNODC, OFAC, coast guard documents
- Objective: model learns to reason across multi-source data and generate maritime intelligence narratives in correct domain language

4.5 Evaluation

Component	Metric	Baseline Comparison
Ship detection	mAP@0.5, Precision, Recall	CFAR (traditional method)
Vessel classification	Per-class accuracy	Zero-shot VLM (no fine-tuning)
Report quality	BERTScore + domain keyword coverage	Human expert evaluation rubric
Prediction accuracy	Position error at 6h/12h/24h (nautical miles)	Simple linear extrapolation baseline

5. Agentic Architecture

The system is built as a multi-agent pipeline orchestrated using LangGraph. Six autonomous agents each perform a specific role and communicate through a shared state object. If any agent fails — for example, AIS data is unavailable for a time window — the pipeline degrades gracefully and notes the data gap in the output report rather than crashing.

5.1 Agent Overview

Agent	Role	Tools Used	Output
SAR Ingestion Agent	Fetches and preprocesses Sentinel-1 imagery	sentinelsat, ESA SNAP / PyroSAR	Preprocessed SAR scene (GeoTIFF)
AIS Correlation Agent	Fetches AIS records, matches to SAR targets, flags dark vessels	Global Fishing Watch API, NOAA AIS	Dark vessel list with gap metadata
Vessel Registry Agent	Cross-matches vessel dimensions to IMO registry, checks sanctions	IMO GISIS, OFAC API, UN list, Equasis	Identity candidates with confidence scores
Inference Agent	Passes fused data to fine-tuned VLM, runs detection + analysis	HuggingFace Transformers, fine-tuned LLaVA	Raw model analysis output
Report Generation Agent	Structures model output into final intelligence report format	LangChain output parser, Jinja2 templates	Formatted investigation report + predictions
Monitoring Agent	Continuously watches for new data, triggers retraining when needed	Airflow, Evidently AI, MLflow, RSS/API monitors	Retraining jobs, sanctions updates, new SAR alerts

5.2 Agent Pipeline Flow

```
USER INPUT: Coordinates + Time Window
    |
    v
+-----+
| SAR INGESTION | Fetches Sentinel-1 pass via Copernicus API
| AGENT         | Applies: calibration > speckle filter >
|               | terrain correction > dB conversion
+-----+-----+
    |
    v
+-----+-----+
| AIS CORRELATION | Fetches AIS records for same area + time
| AGENT           | Spatial-temporal matching of SAR targets to AIS
|                 | Unmatched targets flagged as DARK
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    |
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| VESSEL REGISTRY | IMO registry lookup by estimated dimensions
| AGENT           | OFAC SDN + UN sanctions check
|                 | Ownership chain trace via Equasis
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    |
    v
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```


6. Evolving Pipeline — Self-Updating System

This is what differentiates the system from a static model deployment. The pipeline is designed to improve automatically as the maritime threat landscape changes — new vessel evasion techniques emerge, new sanctions are imposed, new crime patterns are documented.

6.1 What Changes Over Time That the System Must Track

- New vessel evasion techniques — sanctioned actors adapt AIS manipulation methods
- New sanctions — OFAC and UN add vessels and entities on an ongoing basis
- New labeled cases — every UNODC maritime crime report, every coast guard interception is a new training example
- New SAR data — Sentinel-1 produces new passes every 6 days, continuous inference opportunities
- ISRO NISAR (launching 2025) — new L-band SAR covering Indian Ocean every 12 days will be ingested automatically
- New research papers — new detection methods from arXiv and IEEE can be incorporated

6.2 Monitoring Agent Sources

Source	What Is Monitored	Frequency	Action on New Data
UNODC maritime publications	New dark vessel enforcement cases	Monthly	Parse into training pairs, add to pool
OFAC SDN list	New sanctioned vessels and entities	Weekly	Update registry agent database immediately
UN sanctions list	UN Security Council vessel sanctions	Weekly	Update registry agent database
IFC-IOR bulletins	Indian Ocean maritime intelligence	As published	New labeled events for training
ESA Copernicus Hub	New Sentinel-1 passes over monitored zones	Every 6 days	Trigger inference pipeline automatically
arXiv cs.CV / eess.SP	New SAR detection and maritime ML papers	Daily	Flag for team review and incorporation

6.3 Drift Detection and Retraining

- Tool: Evidently AI — open source, Python-native
- Method: statistical comparison of new incoming data features against current training distribution
- Trigger threshold: if Jensen-Shannon divergence between new data and training data exceeds defined threshold → retraining flagged
- Retraining: QLoRA fine-tuning on accumulated new labeled pairs — incremental, not full retraining
- Evaluation: new model evaluated on held-out test set before promotion
- Promotion rule: new model promoted only if it improves on all primary metrics — otherwise queued for next cycle

6.4 Model Registry

- Tool: MLflow — open source, runs locally or on cloud
- Every model version tracked with: training data provenance, training date, evaluation metrics, deployment timestamp

- Full audit trail — at any point it is possible to answer: what did the model know, what data was it trained on, when was it last updated
- Rollback: if a new model version causes degradation in production inference, previous version is restored from MLflow registry

7. Full Technology Stack

7.1 Core Stack

Component	Tool / Library	Purpose
Base Model	LLaVA-1.6 (Mistral-7B) or InternVL2-8B	Vision-language foundation
Fine-tuning	HuggingFace PEFT + QLoRA + bitsandbytes	Domain adaptation
Agent Framework	LangGraph + LangChain	Multi-agent orchestration
SAR Processing	ESA SNAP / PyroSAR / sentinelsat	SAR download and preprocessing
Ship Detection	YOLOv8 fine-tuned on SAR-Ship (Stage 1)	Fast detection before VLM classification
Pipeline Orchestration	Apache Airflow	Scheduling and monitoring agent jobs
Drift Detection	Evidently AI	Data distribution monitoring
Model Registry	MLflow	Model versioning and experiment tracking
Vector Store	FAISS	Semantic search over historical vessel cases
Database	PostgreSQL	Structured storage for AIS records and vessel registry
AIS Data	Global Fishing Watch API + sentinelsat	Vessel tracking data

7.2 Compute Requirements

Task	Hardware	Estimated Time
QLoRA fine-tuning Stage 1	A100 40GB	6-8 hours
QLoRA fine-tuning Stage 2	A100 40GB	4-6 hours
Inference per Sentinel-1 scene	A100 or T4	3-5 minutes
Full pipeline per query	A100	Under 10 minutes
Incremental retraining (new data batch)	A100 40GB	2-4 hours

All training is runnable on Google Colab Pro A100 or university GPU cluster. No proprietary compute required.

8. Research Novelty

The research contribution must be precisely stated. The following table shows what existing systems do and exactly where our system goes beyond them.

System	What It Does	What It Does NOT Do
Global Fishing Watch	AIS gap detection, dark vessel flagging	No SAR fusion, no investigation narrative, no prediction
Sentinel-1 ship detection papers	SAR ship detection (classification)	No AIS correlation, no identity attribution, no report generation
Windward / Spire Maritime	AIS behavioral analysis	Proprietary, no SAR, no narrative output, no prediction
CFAR (traditional)	Basic SAR ship detection	No classification, no downstream analysis, no AIS integration
Our System	All of the above combined end-to-end	First complete pipeline: SAR + AIS + identity + report + prediction

Novel contribution in one sentence: We present the first end-to-end pipeline that fuses SAR imagery with AIS data, performs vessel identity attribution against live sanctions registries, and generates structured maritime investigation reports with predictive trajectory and intercept intelligence using a fine-tuned vision-language model wrapped in a self-updating agentic architecture.

8.1 Publication Target

- Primary: IEEE Transactions on Geoscience and Remote Sensing — Impact Factor 8.2
- Secondary: ISPRS Journal of Photogrammetry and Remote Sensing — Impact Factor 12.7
- Conference: IGARSS 2025 — IEEE International Geoscience and Remote Sensing Symposium

9. Development Roadmap

This roadmap is designed for one academic year. Each phase has clear deliverables and the team can work on phases in parallel where dependencies allow.

Phase	Duration	Tasks	Deliverable
Phase 1 — Data Collection	Weeks 1-4	Set up Copernicus API access, download Sentinel-1 scenes, set up GFW API, download SAR-Ship + HRSID + OpenSARShip datasets, parse UNODC + OFAC documents into training pairs	Complete dataset pipeline, ~50K detection training examples, ~400 report training pairs
Phase 2 — SAR Preprocessing	Weeks 3-6	Build SNAP/PyroSAR preprocessing pipeline, implement calibration + speckle filtering + terrain correction + dB conversion, validate output quality on known ship locations	Automated SAR preprocessing pipeline, validated on 50+ scenes
Phase 3 — Stage 1 Fine-Tuning	Weeks 5-8	Fine-tune YOLOv8 on SAR-Ship for detection, fine-tune LLaVA Stage 1 for classification, evaluate on SSDD benchmark	Ship detector + classifier, mAP results vs CFAR baseline
Phase 4 — AIS Correlation	Weeks 7-10	Build AIS fetching and spatial-temporal matching module, implement dark vessel flagging logic, validate against GFW labeled dark vessel events	AIS correlation module, validated detection rate on labeled events
Phase 5 — Registry and Sanctions	Weeks 9-11	Build IMO/OFAC/UN/Equasis lookup agents, implement vessel dimension matching for identity attribution, build confidence scoring	Registry agent with sanctions integration, tested on historical cases
Phase 6 — Stage 2 Fine-Tuning	Weeks 10-14	Fine-tune LLaVA Stage 2 on report generation pairs, implement prediction module (trajectory, EEZ, intercept), evaluate report quality	Full investigation + prediction generation, BERTScore results
Phase 7 — Agent Integration	Weeks 13-16	Build LangGraph multi-agent pipeline, integrate all components, implement error handling and graceful degradation, end-to-end testing	Full integrated pipeline, end-to-end demo on real Sentinel-1 scenes
Phase 8 — Evolving Pipeline	Weeks 15-18	Build monitoring agent, implement Evidently AI drift detection, set up MLflow model registry, implement Airflow scheduling	Self-updating pipeline, demonstrated on new data batch
Phase 9 — Evaluation + Paper	Weeks 17-22	Full ablation study, quantitative evaluation on all metrics, comparison against baselines, paper writing and submission	Research paper draft, conference or journal submission

10. Quick Reference Summary

Component	Decision
Base Model	LLaVA-1.6 (Mistral-7B) — primary InternVL2-8B — alternative
Fine-tuning Method	QLoRA — two stage (detection then report + prediction generation)
Primary SAR Data	Sentinel-1 — free, global, 6-day revisit, archive since 2014
AIS Data	Global Fishing Watch + NOAA AIS archive
Vessel Registry	IMO GISIS + Equasis
Sanctions Lists	OFAC SDN + UN Consolidated + EU Sanctions
Detection Training Data	SAR-Ship (39K) + HRSID (5.6K) + OpenSARShip (11K)
Report Training Data	~400 pairs constructed from UNODC + OFAC + coast guard documents
Agent Framework	LangGraph — 6 autonomous agents
Pipeline Orchestration	Apache Airflow
Drift Detection	Evidently AI
Model Versioning	MLflow
SAR Preprocessing	ESA SNAP + PyroSAR + sentinelsat Python library
Novel Contribution	First end-to-end SAR + AIS + identity + investigation report + predictive intelligence pipeline
Publication Target	IEEE Transactions on Geoscience and Remote Sensing
Compute Required	A100 40GB — Google Colab Pro or university GPU cluster

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